Weekly Research Summary

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This week

We moved from a broad idea about AI data work to a defined paper on technology shocks and cyber crime against women, with the data assembled and a draft specification.

1. Installed the Supervisor-Skills research toolkit (idea evaluation, literature review, paper-writing skills) in the repo.
2. Evaluated the first idea, South Asian women in AI data-labelling work, and wrote a literature review on it (34 sources). Set it aside.
3. Wrote a literature review on cheap mobile data (Jio) and cyber crime against women (17 sources).
4. Pivoted to the current idea: technology shocks (Jio, the generative-AI boom) and recorded cyber crime against women. Evaluated it: accept with revisions; a strong Master's thesis or field-journal paper.
5. Found all datasets built in earlier sessions across repo branches and listed them; dropped the AI-exposure-of-jobs and PLFS work.
6. Added your internet access workbook (NSS 2017-18, NFHS-5, CAMS 2022-23, TRAI) and switched from gender-split access to total population access.
7. Built three analysis panels (state × year, state × crime category × year, platform × month) with a codebook, and drafted the econometric specification.
8. Merged your Google Trends and Google Ads search data into the panels, including nudify-app searches by state.
9. Gathered every dataset into one `datasets/` folder with a README.
10. Wrote a literature review on econometric analysis of NCRB cyber crime data (14 sources) and checked how states label cyber crimes differently.
11. Made a research inventory doc and saved all notes in `research/` on branch `claude/quirky-meitner-avco1h`.

The idea now

Did technology shocks raise recorded cyber crime against women in India, and did it rise more in states with more internet access? Two shocks made online abuse easier: cheap Jio data (2016) brought far more people online, and generative AI (2022 onwards) made tools such as nudify and deepfake apps widely available.

- **Comparison crimes:** cyber crime against women is compared with offline crimes against women in the same state (same victims and police) and with cyber fraud (online crime not aimed at women). This separates gendered online abuse from general crime growth and from state-wide changes in reporting.
- **Claim level:** associations supported by timing and comparison groups, not causal effects. Outcomes are *recorded* crimes, so a rise may reflect more abuse, more reporting, or both.
- **The AI boom is the main result.** NCRB reports cyber crime against women separately only from 2017, so Jio can only be shown descriptively.
- **Contribution:** the few econometric studies of NCRB cyber crime only ask what predicts crime levels. This would be the first to apply a shock-and-comparison design to NCRB cyber crime, use state internet access, and use the post-2017 tables of cyber crimes against women.

Data we have now

Everything is in the repo's `datasets/` folder, grouped into six sets, all covering 35 State/UT units (boundary changes merged).

Data set	Source	What it covers	Years
Analysis panels	Built from the sets below	State × year (453 rows); state × crime category × year (3,360 rows); platform × month (2,365 rows); codebook	2012–2024; 2017–2024; 2021–2026
NCRB crime	*Crime in India*	Cyber crime against women (total and 7 types), offline crimes against women, cyber fraud, cyber crime by motive, all crime types	2012–2024 (women's cyber from 2017)
Internet access	NSS, NFHS-5, CAMS, TRAI, World Bank	% using the internet by state (NSS 2017-18: 22 states; NFHS-5 2019-21; CAMS 2022-23 incl. age 15–24); TRAI subscriptions per 100 people	Survey years; TRAI 2022–2026

Data set	Source	What it covers	Years
Google Trends and Ads	Google	Trends index for 9 deepfake and face-swap concepts by state; Ads search volumes for 107 keywords incl. nudify terms, by state and month	Trends 2018–2026; Ads Nov 2022–Sep 2026
Platform grievances	IT Rules compliance reports	Monthly user complaints by category: Meta, X, Google, WhatsApp, Snap, Pinterest, ShareChat, Moj, Roblox	2021–2026
Deepfakes and online abuse	Fact-checkers, Reddit, YouTube, text corpora	2,351 deepfake fact-checks; 61 Reddit "fakes" communities; labelled threat-text corpus (255,490 texts, Hinglish subset)	2018–2024

Known limits

- No state data on cyber crime against women or internet access before 2017, so no clean before/after for Jio.
- Nudify search data starts in Nov 2022, so there is no pre-AI-boom nudify measure; pre-boom Trends measures are noisy for small states.
- States label cyber crimes very differently (IT Act share of cases ranges from 2% to 100%; about 60% of cyber crimes against women coded "other" in the typical state; 12 states have large year-to-year jumps).

Possible econometric specification

Three count models estimated by Poisson regression (PPML) with fixed effects, plus a descriptive Jio section. Population enters as an offset so the models explain rates; standard errors are clustered by state, with bootstrap and randomisation-inference checks because there are only 35 states. Notation: *s* state, *c* crime category, *t* year, *m* month.

Model 1: internet access and the growth of cyber crime against women (2017–2024). Does cyber crime against women grow faster than offline crime against women in more-connected states?

```
E[Y_sct] = exp( α_sc + δ_st + λ_ct + β · Cyber_c × Access_s × (t − 2017) + ln
FemalePop_st )
```

- Fixed effects: state × category (each state's usual level of each crime), state × year (anything hitting all crime against women in a state, e.g. police capacity or a new helpline), category × year (national trends, e.g. the 2019 reporting portal).
- Access: total population internet use (CAMS 2022-23 main; NSS 2017-18, NFHS-5, youth 15–24 as alternatives). An event-study version replaces the trend with year dummies.

Model 2: the generative-AI boom (main result). Did image-based cyber crime against women (obscene material, morphing, blackmail) rise more after 2022 in more-exposed states?

```
E[Y_sct] = exp( α_sc + δ_st + λ_ct + β · AI_c × Post_t × Exposure_s + ln
FemalePop_st )
```

- Post = 2023–2024. Exposure options: internet access before the boom (NFHS-5, preferred), nudify-to-general-AI search ratio 2023–24 (closest to the mechanism), nudify searches per person, pre-boom Trends (robustness).
- Event study with 2022 as the base year; flat 2017–2021 coefficients show states were not already diverging.

Model 3: AI image tools on X (national, monthly). Did X's synthetic-media complaints rise after Grok's image tools became free, relative to X's other complaint categories?

```
E[G_cm] = exp( α_c + μ_m + β1 · Synthetic_c × Grok2_m + β2 · Synthetic_c ×
GrokFree_m
              + θ · Synthetic_c × Election_m )
```

- Already estimated earlier (Part A): complaints rose roughly 17–21 times after Grok and Aurora became free, robust to placebo dates.

Jio (descriptive). National internet use and cyber crime trends 2004–2024 with September 2016 marked; no causal claim.

Checks

- Placebos: cyber fraud instead of cyber crime against women; fake shock years.
- Robustness: drop 2020 (COVID) and 2024 (new criminal code); drop high-reporting states (Karnataka, Telangana); log-OLS instead of Poisson; alternative exposure measures.
- Labelling: use total cyber crime against women as the main outcome; flag states with large classification jumps and re-run without them.

Full detail: `research/econometric-specification.md` in the repo.